

Certified Dynamic Reconfiguration of Time-Sensitive Networks: Online Admission Control under Worst-Case Determinism Guarantees

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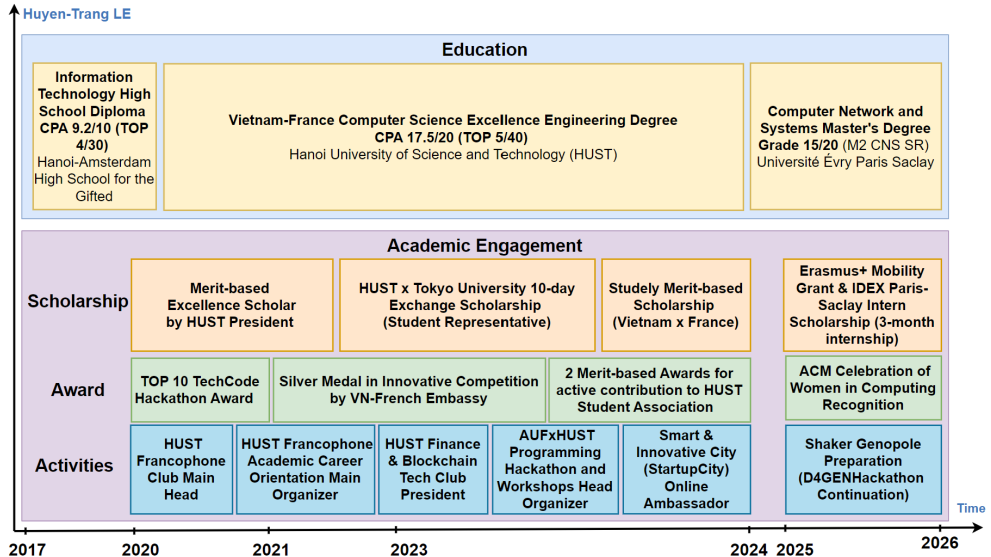
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PhD Thesis Proposal in LyRIDS (ECE) / CEDRIC (Cnam), Paris

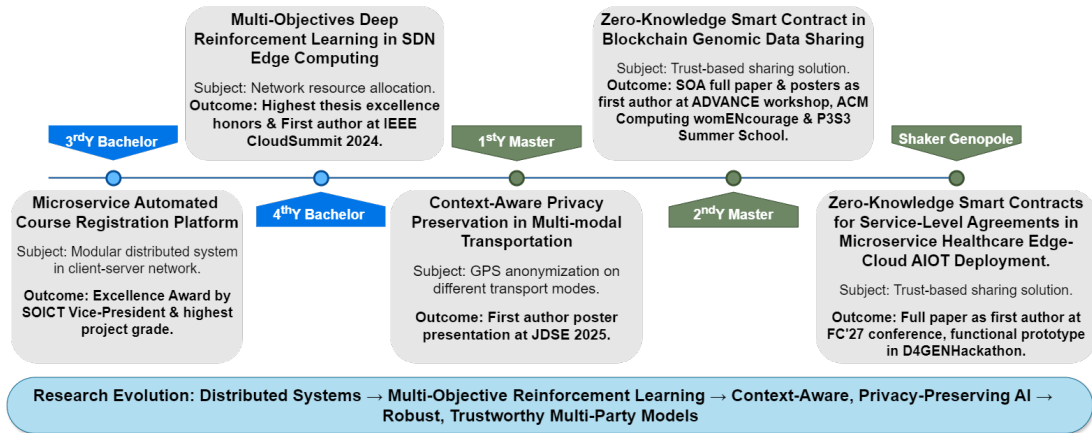
Outline

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1.1 Personal Curriculum Vitae: Education & Academic Engagement



1.2 Personal Curriculum Vitae: Research Experience



2.1 Research Context and Literature Review

From Offline-Configured to Dynamically Reconfigured Deterministic Networks

Safety-critical CPS (automated driving, avionics, Industry 4.0) replace domain-specific fieldbuses with converged **IEEE TSN** Ethernet backbones carrying **mixed-criticality** traffic. IEEE 802.1Qbv (Time-Aware Shaper) and 802.1Qcr (Asynchronous Traffic Shaping) provide **deterministic latency bounds** – but only if flows, routes and resources are **known at design time** and the configuration is **synthesised and verified offline**.

- In operation, **flows appear and disappear**: sensor streams switched on/off, plug-and-produce stations, migrated edge control tasks, software updates.
- Each event is a **reconfiguration request** (admit/reject, reroute, new gate control list, buffer re-dimensioning) to be answered **online without invalidating** the guarantees of flows already in service.
- Configuration is increasingly **automated end-to-end** (zero-touch, multi-domain orchestration, AI agents) [1, 2, 3, 4] $\implies$ **Key Challenges** remain open.

2.1 Research Context and Literature Review

3 Key Challenges (KC) of Certified Online TSN Reconfiguration

Speed vs. Worst-Case Guarantee (KC1)

Exact ILP/SMT synthesis is **NP-hard** and far too slow for reconfiguration deadlines; fast DRL/GNN methods decide in milliseconds [5, 6, 7, 8] but optimise an **expected return**, not a **certified timing envelope**.

Online Verification under a Deadline (KC2)

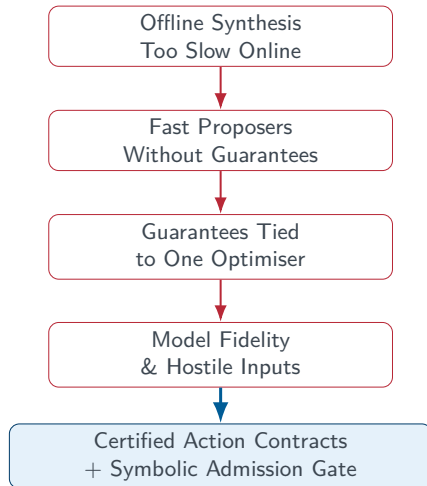
Worst-case analysis by **deterministic network calculus** [9, 10] must be re-done **within the reconfiguration deadline**, on industrial-scale topologies.

Boundary of the Guarantee (KC3)

A certificate is a proof **about a model**: stale telemetry, corrupted flow database or **gPTP attacks** [11] break the link between model and network.

2.1 Research Context and Literature Review

- **Learning-based configuration:** DRL/GNN/MARL policies for TSN admission, routing and scheduling of incremental flows [5, 6, 7, 8]; GNN delay prediction enhanced by network calculus [12] in **no worst-case certificate**.
- **Dominant answer: constrain the optimiser**, shielding [13], probabilistic shielding [14]: the guarantee is tied to **one optimiser**, a fixed action space, and gives the operator **no inspectable artefact**.
- **Thesis dual approach:** keep the optimiser fast, unconstrained and **interchangeable**; move the guarantee into a **certified admission stage** owned by the operator.



2.2 Key Definitions (1/2): Deterministic Network Calculus

Arrival and service curves

Flows [9] are wide-sense increasing cumulative functions R (input), R^* (output); min-plus convolution $(f \otimes g)(t) = \inf_{0 \leq s \leq t} \{f(t-s) + g(s)\}$.

Arrival curve α : $R(t) - R(s) \leq \alpha(t-s) \forall s \leq t$ ($R \leq R \otimes \alpha$), e.g. token bucket $\gamma_{r,b}(t) = rt + b$.

Service curve β : $R^* \geq R \otimes \beta$, e.g. rate-latency $\beta_{R,T}(t) = R[t-T]^+$.

Worst-case bounds

Delay $\leq h(\alpha, \beta) = \sup_{t \geq 0} \inf\{d \geq 0 : \alpha(t) \leq \beta(t+d)\}$; **backlog** $\leq v(\alpha, \beta) = \sup_{t \geq 0} \{\alpha(t) - \beta(t)\}$.

For $\gamma_{r,b}$, $\beta_{R,T}$, $r \leq R$: $d \leq T + b/R$, backlog $\leq b + rT$, output arrival curve $\gamma_{r,b+rT}$.

Concatenation: a path offers $\beta_1 \otimes \beta_2 \otimes \dots$ (rate-latency: $\beta_{\min_i R_i, \sum_i T_i}$) in “pay bursts only once” [9].

Application to TSN

Per-class service curves for **credit-based shaped (AVB)** traffic [10], accounting for the **TAS** windows of scheduled traffic $\Rightarrow$ **worst-case end-to-end delay bounds**.

2.2 Key Definitions (2/2): TSN & Shielding

TSN configuration objects

Gate Control List (GCL, 802.1Qbv): cyclic per-port schedule opening/closing the gate of each egress queue; **ATS (802.1Qcr)**: asynchronous, per-flow shaping without a global schedule; **802.1Qcc**: stream reservation and (centralised) configuration interfaces through which flows are requested and configurations are deployed; **gPTP**: time synchronisation on which scheduled traffic relies [11].

Shielding

A **shield** [13, 14], synthesised from a safety specification and an abstraction of the environment, monitors the agent and **blocks or corrects unsafe actions** (before or after the agent's choice), guaranteeing safety during learning and execution; **probabilistic shielding** extends it to probabilistic safety constraints.

Positioning

Shielding binds the guarantee to the learning loop. Here, the “shield” becomes a **network-level admission gate** whose decisions come with a **machine-checkable certificate**, independent of the proposer (heuristic, ILP/SMT, DRL/GNN, LLM/agentive configurators [3, 4]).

2.2 Problem Formulation (1/2): Certified Action Contract

Proposed formulation (built on [9, 10]; to be refined in O1–O2)

- **Runtime state** $S = (\mathcal{G}, \mathcal{F}, \mathcal{C}, \Theta)$: topology $\mathcal{G} = (V, E)$, admitted flows $\mathcal{F}$, configuration $\mathcal{C}$ (routes π_f , GCLs, shaper parameters, buffer sizes), telemetry Θ .
- Each flow f : arrival curve α_f , path π_f , deadline D_f , jitter envelope J_f ; per-hop service curves $\beta_{f,h}(S)$ induced by $\mathcal{C}$.
- **Timing invariant**:

$$\text{Inv}(S) \equiv \forall f \in \mathcal{F} : h\left(\alpha_f, \bigotimes_{h \in \pi_f} \beta_{f,h}(S)\right) \leq D_f \wedge \text{jit}_f(S) \leq J_f \wedge \forall q : v(\alpha_q, \beta_q(S)) \leq B_q$$

- **Actions** $a \in \mathcal{A}$: admit, reroute, resynthesise GCL, remap class, resize queue; $S' = a(S)$.
- **Certified action contract** $\langle \text{Pre}_a, \text{Post}_a \rangle$ with $\text{Post}_a \Rightarrow \text{Inv}(S')$, and **certificate** c_a (affected flows, per-hop bounds, SMT witness) such that an **independent checker** $\text{Check}(S, a, c_a)$ validates it.

Rule

An action **without a valid certificate is never committed** to the data plane.

2.2 Problem Formulation (2/2): Online Gate & Soundness

- **Admission gate** with reconfiguration deadline τ :

$$\Gamma(S, a, \tau) \in \{ \text{Commit}(a, c_a), \text{Fallback}(a_{\text{safe}}, c_{\text{safe}}), \text{Reject} \}, \quad T_{\text{ver}}(a) \leq \tau$$

where fallbacks are a conservative admission, a pre-verified safe configuration, or an explicit refusal.

- **Incrementality**: re-verify only flows $\mathcal{F}_a \subseteq \mathcal{F}$ whose paths cross the sub-graph $\mathcal{G}_a$ affected by a , reusing previously established bounds.
- **Target soundness result (safety)**: for *any* proposer $\mathcal{P}$ and *any* request sequence,

$$\text{Inv}(S_0) \implies \forall k \geq 0 : \text{Inv}(S_k) \quad \text{for the states reached through } \Gamma \circ \mathcal{P}.$$

- **Not guaranteed**: liveness / QoS (starvation, thrashing, low utilisation). **Model fidelity**: the proof holds for the model S , not for a diverging real network $\hat{S}$ (stale telemetry, corrupted flow database).

2.3 Scientific Objectives (O)

Enable Dynamic TSN Reconfiguration while Formally Guaranteeing Worst-Case Determinism.

- **O1 – Runtime network model & action space (KC2):** formalise the runtime state of a TSN domain and the 802.1Qcc-style reconfiguration actions, with their explicit assumptions.
- **O2 – Certified action contracts (KC1):** pre/post-conditions on worst-case envelopes, semantics, composition along paths and across successive actions, certificate content and independent re-checking.
- **O3 – Online verification under a deadline (KC2):** NC + SMT procedure, incremental on the affected sub-graph, anytime with certified fallbacks, soundness proof; bound tightness vs. time budget.
- **O4 – Proposers & cost of certification (KC1):** heuristics, incremental ILP/SMT, DRL/GNN inside the certified envelope, gate verdicts as feedback; optimality gap, admission/rejection ratio, latency.
- **O5 – Evaluation, scalability & robustness (KC3):** formal analysis, OMNeT++/INET TSN and ns-3 simulation, testbed; degraded telemetry, stale flow database, gPTP manipulation, adversarial request sequences.

2.4 Research Questions (RQ) & Methodological Approaches (MA)

RQ1: How to formally express reconfiguration actions and their timing guarantees as contracts whose certificates can be checked independently of the proposer (O1, O2)?

- **Runtime Model & Action Space (MA1):** formal TSN state, 802.1Qcc-style actions, explicit assumptions per action.
- **Contract Language (MA2):** pre/post-conditions over arrival/service curves, per-hop budgets, deadlines, buffer and jitter envelopes; compositional reasoning along paths [9, 10].

RQ2: How to discharge contracts online within the reconfiguration deadline while keeping a soundness guarantee (O3)?

- **Incremental NC + SMT Verification (MA3):** affected sub-graph, reuse and monotonicity of established bounds.
- **Anytime Gate & Soundness (MA4):** certified degraded fallbacks, proof that $\Gamma \circ \mathcal{P}$ preserves Inv for any $\mathcal{P}$ in the dual of shielding [13, 14].

2.4 Research Questions (RQ) & Methodological Approaches (MA)

RQ3: What is the cost of certification, and how robust is the gate at the boundary of its assumptions (O4, O5)?

- **Proposers inside the Envelope (MA5):** constructive heuristics, warm-started ILP/SMT, DRL/GNN policies [5, 6, 7, 8] and zero-touch/agentic configurators [1, 3, 4], using gate verdicts as a feedback signal; reproducible baselines [15].
- **Robustness Campaign (MA6):** degraded/inconsistent telemetry, stale flow database, gPTP-level manipulation [11], adversarial request sequences (starvation, thrashing), building on TSN digital-twin security testing [16].

Expected contributions

Formal notion of certified action contract with a soundness result · incremental online NC-based verification with a characterised tightness/time trade-off · quantified cost of certification across proposer classes · open-source certified TSN reconfiguration control plane and reproducible benchmark.

3. PhD Work Plan: 3-Year Workflow

	Year 1 (M1–M12) SOA, Model, Contracts & First Gate			Year 2 (M13–M24) Full Action Space & Online Verification			Year 3 (M25–M36) Scalability, Robustness & Writing		
Task	11/2026	03/2027	07/2027	11/2027	03/2028	07/2028	11/2028	03/2029	07/2029
Literature Review & Runtime Model	TSN synthesis, admission control, NC, runtime verification; simulation setup								
Contract Language & First Admission Gate		Contracts + gate on flow admission (offline verification baseline)							
Incremental Online Verification				Full action space, NC+SMT, anytime & certified fallbacks					
Proposers & Cost of Certification					Heuristic, ILP/SMT, DRL/GNN proposers; gate feedback				
Scalability & Robustness Campaign							Industrial topologies, testbed, degraded & adversarial inputs		
PhD Defense								Manuscript & defense (10/2029)	
Summer Schools & Academic Activities			Real-time / networking summer schools				Open-source certified control plane & benchmark		
Scientific Perspective		Conference paper: contract model & gate			Journal paper: certified online reconfiguration			Journal paper: scalability & robustness	

Analyze
 Design
 Develop
 Review/Test
 Contribute

4. Conclusion: Motivation to Pursue a PhD

Pathway to Academic Career

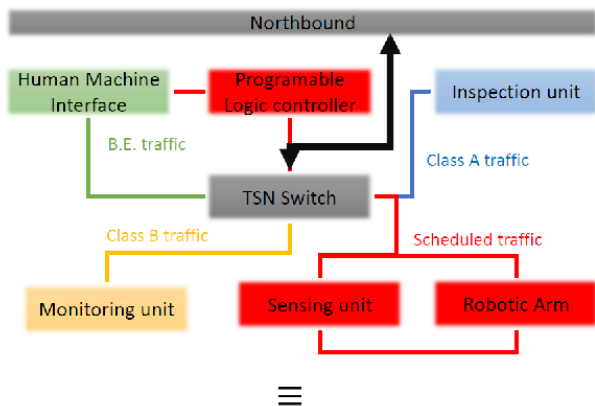
My passion for science, perseverance, and strong motivation.

- **Inspiration:** from **Aakash Soni** and **Samia Saad-Bouzefrane**, whose work at **LyRIDS (ECE)** and **CEDRIC (Cnam)** connects **deterministic networking**, **real-time systems** and **secure, trustworthy networked systems**.
- **Goal:** to become a **Researcher and Lecturer** contributing to **scientific progress**, **training and supporting students** (in particular those facing challenges similar to mine as I was supported once by my professors), and promoting **Young Women in Science**.
- **Motivation:** my late mother (university professor) and **Vietnamese professors** who provided me **financial & educational support** for my graduate studies.

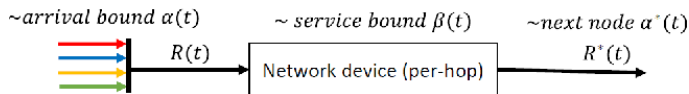
Thank you for your attention!

I am open to your questions

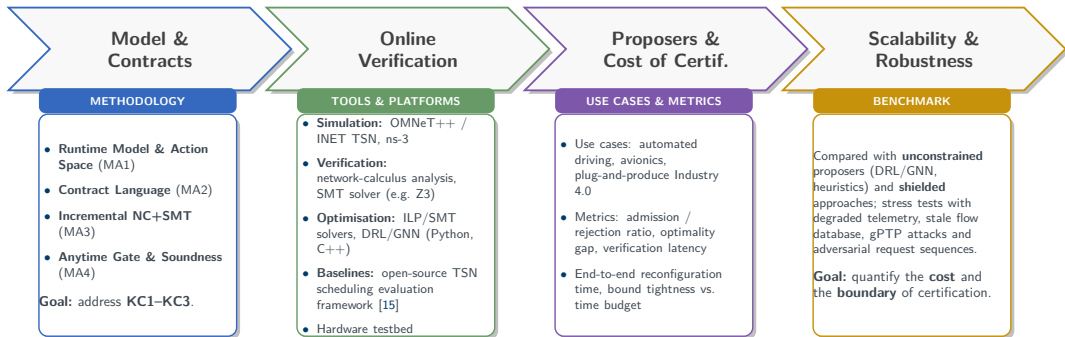
Conceptual Architecture



An **uncertified proposal layer** (heuristic, ILP/SMT, DRL/GNN, zero-touch/agentive configurators) produces candidate actions a . A small, auditable **symbolic admission gate** checks each action's contract against the **runtime model** using network calculus and SMT, and only **certified** actions reach the data plane. Verdicts are fed back to proposers; telemetry keeps the model up to date within its **fidelity** bounds the guarantee.



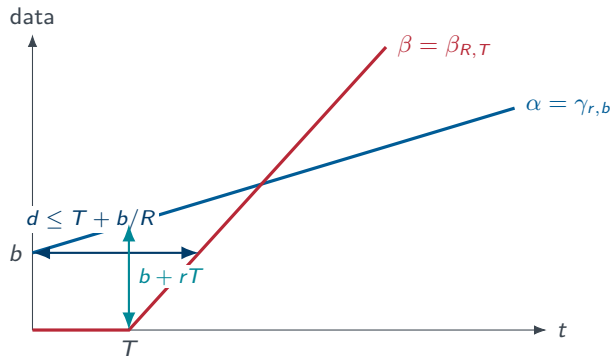
Working Method



Underlying Concepts: Network Calculus Bounds

Delay and backlog as deviations between curves

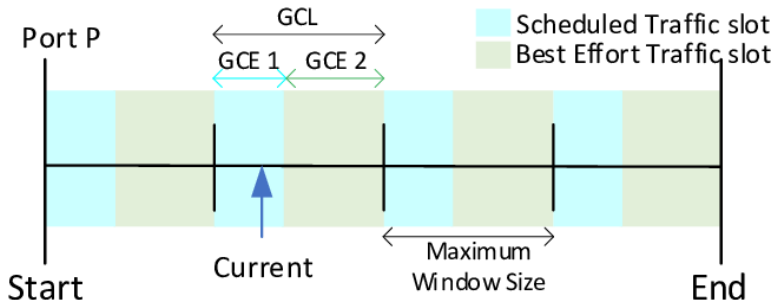
The worst-case delay [9] is the **horizontal** deviation and the worst-case backlog the **vertical** deviation between the arrival curve α and the service curve β .



Underlying Concepts: TSN Shaping & Scheduling

Time-Aware Shaper, ATS and CBS

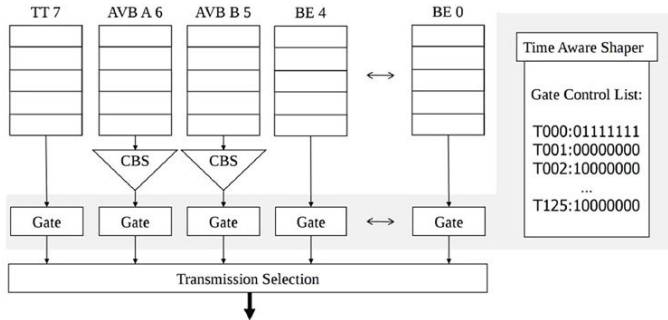
The TAS opens and closes queue gates according to a cyclic **GCL** [10]; scheduled traffic gets protected windows, while shaped classes (CBS/AVB, ATS) receive a service curve reduced by these windows and guard bands.



Underlying Concepts: Shielding vs. Admission Gate

Two places to put the guarantee

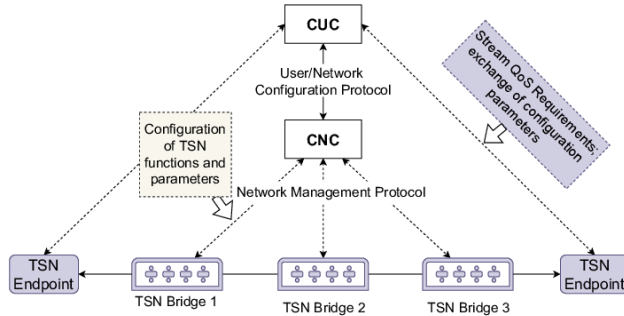
Shielding [13, 14] constrains *one* learning agent's actions using a safety specification. The **certified gate** checks *any* proposed action against network-level contracts and emits a certificate the operator can inspect and re-check.



Underlying Concepts: Online Admission Sequence

Anytime verification with certified fallbacks

Request $\rightarrow$ proposal $a \rightarrow$ affected sub-graph $\mathcal{G}_a \rightarrow$ incremental NC + SMT check $\rightarrow$ commit with c_a , or, if the deadline τ is reached, conservative admission / pre-verified safe configuration / explicit refusal.



Underlying Concepts: Zero-Touch TSN Configuration

Why the gap is structural

Zero-touch management, multi-domain orchestration, LLM-based and agentic configurators [1, 2, 3, 4] automate the configuration path end-to-end: the certified gate is the **operator-owned** safety point in this pipeline.

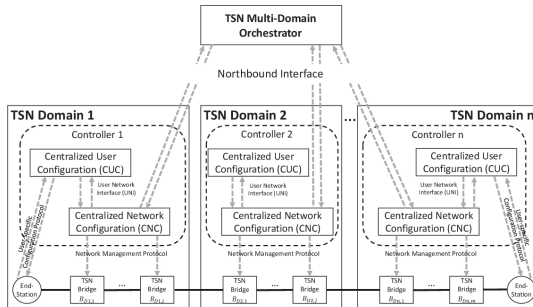
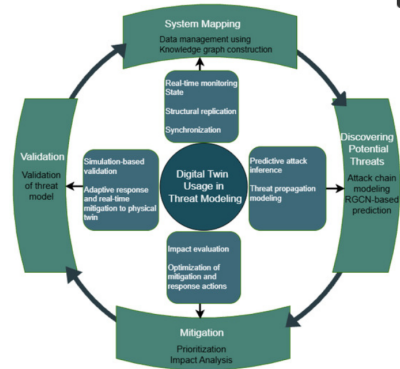


Fig. 2: Proposed architecture for multi-domain TSN

Underlying Concepts: Robustness at the Boundary

A certificate is a proof about a model

Stress scenarios: degraded or inconsistent telemetry [11, 16], stale flow database, clock-synchronisation degradation and **gPTP manipulation**, adversarial request sequences inducing **starvation** or **thrashing**; STRIDE-based testing on a TSN digital twin as a starting point.

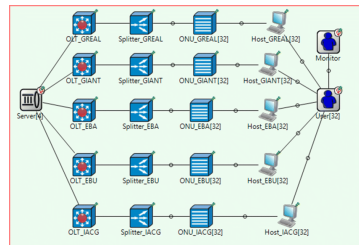


Underlying Concepts: Evaluation Environment

Simulation, formal analysis and testbed

Resources used to **build**, and **evaluate** the certified reconfiguration control plane.

- **Simulation:** OMNeT++, ns-3.
- **Baselines:** TSN scheduling evaluation framework [15]; DRL/GNN schedulers [5, 6, 7].
- **Testbed:** TSN hardware when available (LyRIDS / CEDRIC).



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